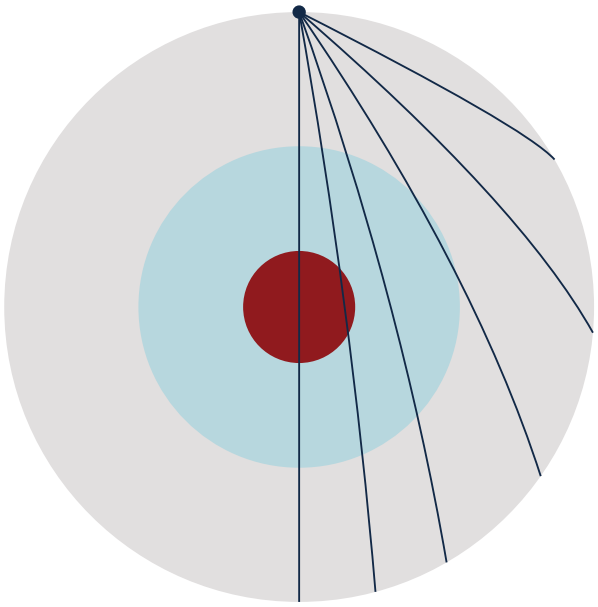
KØBENHAVNS UNIVERSITET

DET NATUR- OG BIOVIDENSKABELIGE FAKULTET
INSTITUT FOR MATEMATISKE FAG



SPECIALE

# Om jordens indre kerne

Et eksempel på en afhandling skrevet med ucph-lehmann-thesis

Inge Lehmann abc123

Vejleder: Prof. Niels Erik Nørlund
Afleveret: 15. oktober 2026

## Oplysninger

| | |
|---|---|
| **Fakultet** | Det Natur- og Biovidenskabelige Fakultet |
| **Institut** | Institut for Matematiske Fag |
| **Forfatter** | Inge Lehmann (abc123) |
| **Titel** | Om jordens indre kerne |
| **Undertitel** | Et eksempel på en afhandling skrevet med ucph-lehmann-thesis |
| **Beskrivelse** | Et eksempel, der viser skabelonens funktioner. |
| **Vejleder** | Prof. Niels Erik Nørlund |
| **Afleveret** | 15. oktober 2026 |
| **Anslag (inkl. mellemrum)** | 5098 (2.1 normalsider) |

# Resumé

Dette dokument er et eksempel på skabelonen `ucph-lehmann-thesis` til Københavns Universitet. Det viser forsiden med det officielle logo, oplysningssiden, der dannes ud fra afhandlingens metadata, resuméer på to sprog, indholdsfortegnelsen og konventionerne for kapitler, figurer, tabeller, ligninger, referencer og appendiks.

# Abstract

This document is an example of the `ucph-lehmann-thesis` template for the University of Copenhagen. It shows the front page with the official logo, the information page generated from the thesis metadata, abstracts in two languages, the table of contents, and the conventions for chapters, figures, tables, equations, references and appendices. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At.

# Forord

The template is named after Inge Lehmann (1888–1993), who graduated from the University of Copenhagen in 1920 and in 1936 showed that the Earth has a solid inner core. Her paper announcing the discovery carried what may be the shortest title in the history of geophysics: “P′” (Lehmann, 1936). Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

# Indhold

**Resumé** ii

**Abstract** iii

**Forord** iv

**1 Introduction** 1

1.1 Motivation . . . . . 1

1.2 Structure of the thesis . . . . . 1

**2 Using the template** 2

2.1 Fonts . . . . . 2

2.2 Characters and normal pages . . . . . 2

**3 Results** 3

3.1 A figure . . . . . 3

3.2 An equation . . . . . 3

3.3 A table . . . . . 4

**4 Conclusion** 5

**Litteratur** 6

**A Additional material** 7

# Figurer

Figur 3.1 A cross-section of the Earth with the inner core in red. Rays from an earthquake at the top bend as they pass through layers of different density. . . . . 3

# Tabeller

Tabel 3.1 Illustrative regression estimates. Standard errors in parentheses. . . . . 4
Tabel 1.1 Approximate layers of the Earth. . . . . 7

# 1 Introduction

A thesis is a long document, and most of its typography should be decided once and then forgotten. This template takes care of the University of Copenhagen front page, the page numbering, and the numbering of chapters, figures, tables and equations, so that the author can concentrate on the content.[1]

The front matter uses roman page numbers and the main text starts again at page 1. Figures and equations are numbered within chapters, as in Figur 3.1 and Ligning (3.1). References are written with `@key` and set in author–year style, e.g. Boserup (1965) or (Schelling, 1971).

## 1.1 Motivation

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

## 1.2 Structure of the thesis

Kapitel 2 describes the setup, Kapitel 3 presents the results and Kapitel 4 concludes. Additional material is collected in Appendiks A.

[1] Footnotes are counted towards the number of characters, as are headings and the body text.

# 2 Using the template

The whole template is configured in one show rule at the top of the document:

```
#import "@preview/ucph-lehmann-thesis:0.1.0": thesis, appendix

#show: thesis.with(
  lang: "en",                 // or "da"
  type: [Master's thesis],
  title: [On the Inner Core of the Earth],
  authors: ((name: "Inge Lehmann", id: "abc123"),),
  advisors: ("Prof. Niels Erik Nørlund",),
  date: datetime(year: 2026, month: 10, day: 15),
  faculty: [Faculty of Science],
  department: [Department of Mathematical Sciences],
  abstracts: (en: [...], da: [...]),
  character-count: true,
)
```

## 2.1 Fonts

The body is set in Liberation Serif and the headings in Liberation Sans, which have the same metrics as Times New Roman and Arial. If they are not installed, the template falls back to Times New Roman/Arial and finally to Libertinus Serif, which is built into Typst.

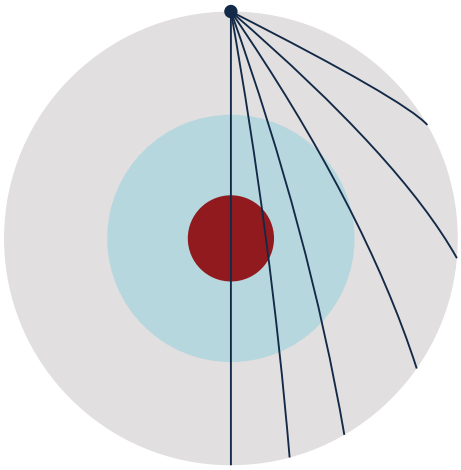
## 2.2 Characters and normal pages

At UCPH, a normal page is 2,400 characters including spaces. With `character-count: true`, the information page reports the number of characters in the main text, stopping at the appendix and leaving out figures, tables, equations and the reference list. Rules differ between programmes, so check yours.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

# 3 Results

## 3.1 A figure



**Figur 3.1.** A cross-section of the Earth with the inner core in red. Rays from an earthquake at the top bend as they pass through layers of different density.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At.

## 3.2 An equation

The ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}, \tag{3.1}$$

which is consistent under the usual assumptions (Angrist og Pischke, 2009). Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

## 3.3 A table

**Tabel 3.1.** Illustrative regression estimates. Standard errors in parentheses.

| | (1) | (2) | (3) |
|---|---|---|---|
| Treatment | 0.124 | 0.118 | 0.097 |
| | (0.031) | (0.030) | (0.028) |
| Controls | No | Yes | Yes |
| Fixed effects | No | No | Yes |
| Observations | 12,408 | 12,408 | 12,408 |

Tabel 3.1 shows how a regression table looks with the default settings. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

# 4 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

# Litteratur

Angrist, J.D. og Pischke, J.-S. (2009) *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton University Press.

Boserup, E. (1965) *The Conditions of Agricultural Growth: The Economics of Agrarian Change under Population Pressure*. London: George Allen & Unwin.

Lehmann, I. (1936) “P'”, *Publications du Bureau Central Séismologique International, Série A, Travaux Scientifiques*, 14, s. 87–115.

Schelling, T.C. (1971) “Dynamic models of segregation”, *Journal of Mathematical Sociology*, 1(2), s. 143–186.

# A Additional material

Appendix chapters are lettered, and so are their figures, tables and equations:

$$y_i = \alpha + x_i\beta + \varepsilon_i \tag{A.1}$$

**Tabel A.1.** Approximate layers of the Earth.

| Layer | Depth (km) | State |
|---|---|---|
| Crust and mantle | 0–2,900 | Solid |
| Outer core | 2,900–5,150 | Liquid |
| Inner core | 5,150–6,370 | Solid |

Ligning (A.1) and Tabel A.1 are numbered within the appendix.